

The marginal value of a model over the market

Paper 4 in a series on UK All-Weather racing outcomes

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2026-09-03

Papers 1 to 3 fitted three models to UK All-Weather Flat handicaps and the market outpredicted all three. This paper asks whether the model adds anything to the price rather than whether it beats it, using a two-stage conditional logit (Benter 1994) in which a market probability and a model probability enter with both coefficients free. It reuses the stored test-set predictions of papers 2b and 3; neither model is refitted. Against the industry starting price, pooled over 2,182 held-out races, the model coefficient is 0.068 and 0.039, with 95% intervals reaching no higher than 0.204. Against the pre-race racecard forecast price it is 1.04 and 1.03 on the earlier half of the test window and 0.75 and 0.72 on the later half. Against the settled market the model adds nothing distinguishable. Against the morning forecast price it adds a great deal. The two results together locate the model's value in the interval between an early price and the off, rather than in information the settled market lacks.

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1 Introduction

Papers 1 to 3 fitted three models to UK All-Weather Flat handicaps — a conditional logit on Owen’s features (Owen 2019), the same model on an extended feature set and a ranking objective, and a gradient-boosted tree ensemble — and the market outpredicted all three. This paper asks a different question of the same predictions: not whether the model beats the price, but whether it adds anything to it.

The construction is a two-stage conditional logit (Benter 1994). A market probability and a model probability enter a race-conditional logit as regressors with free coefficients, and the coefficient on the model term, β_{mod} , is the output. If the price already carries everything the model knows, β_{mod} is zero. If the model holds information the price does not, it is positive. Papers 1 to 3 cannot measure this: a model that is strictly dominated and a model that is partly complementary both lose the head-to-head.

The series kept market data out of the feature set so that each change could be read against the last. Neither price is leakage: the forecast price is published the evening before a race, and the series has used starting prices all along to settle its backtests. Here the model still uses no market data — it is the evaluation that sees the price.

Papers 2a to 3 changed the feature set, the objective and the function class in turn, and none closed the gap to the market. This paper changes none of them. It comes before the representation learning paper 3 pointed toward because it sets that work’s bar: if the settled market already prices what these features carry, a learned representation of the same run histories has to create information rather than encode it better.

There is no betting-value question here, unlike papers 1 to 3. A blend that tracks the starting price bets into the starting price’s own margin, so the backtest has a foregone conclusion.

2 Method

2.1 The two-stage conditional logit

For runner i in race j ,

$$V_{ij} = \beta_{\text{mkt}} \log P_{ij}^{\text{mkt}} + \beta_{\text{mod}} \log P_{ij}^{\text{mod}}, \quad \Pr(i \text{ wins race } j) = \frac{e^{V_{ij}}}{\sum_{k \in j} e^{V_{kj}}}, \quad (1)$$

where P^{mkt} is a market-implied win probability and P^{mod} a model win probability. Benter used this to combine a fundamental model with the public odds (Benter 1994). Race is the choice set, as in every conditional logit in this series, and the fit is by maximum likelihood over the observed winners. There is no intercept: an alternative-invariant constant cancels in the softmax, so the three-part `{mlogit}` formula `won ~ ... | 0 | 0` used in papers 1 to 2b carries over unchanged.

Both probabilities are already normalised within race, so Equation 1 is a weighted geometric combination of them, renormalised. A coefficient of 1 uses a source as stated; above 1 sharpens it toward the favourite, below 1 flattens it. β_{mod} is the output.

2.2 Why β_{mkt} is not pinned at 1

Fixing $\beta_{\text{mkt}} = 1$ would assert that the market’s probabilities need no sharpening or flattening before use. If that is wrong, the likelihood has only one free parameter left to correct it with, and part of β_{mod} becomes a recalibration of the market rather than a measure of the model. The bias runs upward, and worst where the market is worst calibrated — which is where a large β_{mod} would otherwise look most like a finding. Both coefficients are therefore free.

In the two starting-price blends, one per model source, the market coefficient comes out at 1.090 and 1.102, close to 1.

2.3 Three prices

Table 1: The three prices.

Arm	Price	What it is
A	Pre-race racecard forecast	The forecast price as published ahead of the race, from Smartform’s daily feed of forthcoming racecards. Available at bet time.
B	Industry starting price	The settled market at the off, and the price the series already uses for its market baseline and its backtests.
C	Archived racecard forecast	The forecast price as recorded in Smartform’s results archive. Nominally the same quantity as arm A, and not identical to it (Section 7).

Arms A and B are the contrast the paper is about: an early price against the settled one. Arm C exists because the archived value is the obvious one to reach for and does not match the pre-race one; it supports the attenuation bracket in Section 5. Section 7 gives the schema detail and the evidence.

All three arms are fitted on one race set — the intersection over the three prices — because a comparison of β_{mod} across arms is void if the arms sit on different races.

2.4 Market probability construction

Each price becomes a win probability by the proportional overround adjustment the series has used since paper 1 §5.1: implied probability $1/\text{price}$ per runner, renormalised to sum to one over the field. Paper 4 does not restate that arithmetic in code; the function it calls is checked against the series’ existing implementation by a standing gate, which rebuilds the stored `win_market` column of paper 3’s test predictions from raw starting prices and requires agreement to 10^{-12} . Observed agreement is exact.

Normalisation is over the runners that started, which is the field the model scores. The forecast price is struck against the declared field, before withdrawals are known, so a race that loses runners shows a lower overround here than the one the bookmaker struck. Renormalisation is unaffected: it rescales to sum to one either way.

Races are dropped whole where any runner lacks a usable price; dropping individual runners would change the normalisation for the rest of the field. Both probabilities are floored at 10^{-6} before taking logs.

2.5 Two model sources, neither refitted

P^{mod} comes from paper 2b’s exploded conditional logit and paper 3’s gradient-boosted tree ensemble, read from the stored test-set predictions of their own papers. Neither is refitted or retuned. The two sit on identical runner sets over identical races, which is what makes the comparison between them paired.

2.6 Where the blend is fitted

β_{mod} is only meaningful on out-of-sample model probabilities. Papers 2b and 3 were fitted on the training split, so their training-split probabilities are optimistic and the blend would read that optimism as information the market lacks. The blend therefore sits on the held-out races — the one place this paper sets aside the series’ training-split-only convention, which Section 3 keeps. A two-parameter model on roughly a thousand races carries negligible overfitting risk; the same could not be said of the models supplying P^{mod} .

The test universe is split chronologically in two. Test-A, the earlier half, is fitted first; test-B was held back. The boundary is a calendar date, so the split is reproducible against an exact date, as the series’ 2012-12-30 train/test cutoff is; races on the cutoff date go to test-A. Section 4 reports what test-B was spent on.

3 Data

The race universe is unchanged from papers 2b and 3: Class 2–5 All-Weather Flat handicaps at Kempton, Lingfield, Southwell and Wolverhampton, 2006-01-01 to 2015-10-14, non-maiden, four to sixteen starters, with the same chronological 70/30 train/test split at 2012-12-30. Paper 1 sets out the scope decisions. Only the price data is new, and only that is covered here. Everything distributional below is training-split; coverage is reported on both splits.

3.1 Coverage

A price is usable if present and strictly greater than 1. The pre-race forecast additionally requires its feed row to have been written before the meeting date.

Table 2: Usable-price coverage by year, over all in-scope runner rows.

Year	Races	Runner rows	Archived forecast	Pre-race forecast	Starting price
2006	635	6,404	87.1%	0.0%	99.8%
2007	685	6,472	97.6%	0.0%	100.0%
2008	728	6,427	100.0%	25.6%	100.0%
2009	839	7,543	100.0%	94.2%	100.0%
2010	819	7,322	100.0%	94.9%	100.0%
2011	690	6,201	100.0%	97.7%	100.0%
2012	816	7,548	100.0%	98.4%	100.0%
2013	830	7,074	100.0%	100.0%	100.0%
2014	872	7,347	100.0%	100.0%	100.0%
2015	527	4,432	100.0%	99.9%	100.0%

The two forecast prices fail in opposite places. The archived value is complete from 2008 and thin only in 2006, at 87.1%. The pre-race value does not exist before March 2008, when Smartform’s daily feed begins, so training-split coverage is 60.9%. On the test split, where this paper fits, it is 100.0%. The pre-race value is therefore not a drop-in replacement for the archived one elsewhere in the series.

Races are dropped whole, so race-level completeness governs the fit.

Table 3: Race-level completeness over the test universe, and the intersection every arm is fitted on.

Price set	Races complete	Of	%	Lost
A — pre-race racecard forecast	2,182	2,183	99.95%	1
B — industry starting price	2,183	2,183	100.00%	0
C — archived racecard forecast	2,183	2,183	100.00%	0
intersection (all three)	2,182	2,183	99.95%	1

The intersection costs one race, lost to the pre-race forecast. Every arm is fitted on the remaining 2,182.

3.2 Overround and shape

Table 4: Race overround (sum of $1/\text{price}$ over the field) by price source. Training split, and the 2,788 races carrying a complete set of all three prices — the pre-race forecast does not exist before March 2008, so putting the three on their own larger sets would not compare like with like.

Source	Races	25th pct	Median	75th pct	IQR
Pre-race forecast	2,788	1.086	1.150	1.201	0.115
Archived forecast	2,788	1.100	1.160	1.206	0.105
Starting price	2,788	1.126	1.156	1.191	0.065

Median margins are similar — 1.160 for the archived forecast against 1.156 for the starting price — but the forecast prices are more dispersed, with an interquartile range of 0.105 against 0.065. The margin also widens with field size, from 1.053 at 4 runners to 1.288 at 14, the largest field with enough races in this set — a gradient 69% steeper than the starting price’s.

Regressing the within-race centred log archived-forecast probability on the within-race centred log starting-price probability gives a slope of 0.554 (95% CI [0.549, 0.559], 46,614 runners in 5,081 training races). Below 1 means the forecast price separates the field less than the starting price does: it spreads runners over a narrower range of probabilities, making favourites too long and longshots too short.

4 Results

The blend is fitted on the 2,182-race intersection (Section 3), split chronologically at 19 March 2014 into test-A (1,095 races) and test-B (1,087 races). The probability floor binds on 0 rows.

Test-A was fitted first, with test-B untouched. The starting-price result below then made test-B’s reserved purpose — evaluating blend performance — moot, and it was spent instead on replicating β_{mod} at the identical specification. That leaves no held-out race set in the paper-4 universe. One thing touched test-B beforehand: the two-price diagnostic in Section 7, which by design runs on the whole test universe. It contains no model probability, so it cannot leak model performance, and it changed no arm.

4.1 The starting price

Table 5: The model coefficient when the market term is the starting price. Test-A and test-B are disjoint halves of the test universe; pooled is both together.

Model	Races	Sample	β_{mod}	SE	95% CI	z
Paper 2b	1,095	Test-A (earlier)	0.152	0.098	[-0.040, 0.343]	1.55
Paper 2b	1,087	Test-B (later)	-0.015	0.098	[-0.207, 0.177]	-0.15
Paper 2b	2,182	Pooled	0.068	0.069	[-0.067, 0.204]	0.99
Paper 3 (GBT)	1,095	Test-A (earlier)	0.124	0.100	[-0.071, 0.320]	1.24
Paper 3 (GBT)	1,087	Test-B (later)	-0.040	0.096	[-0.229, 0.149]	-0.42
Paper 3 (GBT)	2,182	Pooled	0.039	0.069	[-0.097, 0.175]	0.56

Pooled, β_{mod} is 0.068 for paper 2b and 0.039 for paper 3, neither distinguishable from zero. The intervals reach no higher than 0.204, which is 28% of the smallest coefficient the same models produce against the forecast price.

The halves agree, which licenses the pooling: the difference between them is not distinguishable from zero for either model ($p = 0.23$ and 0.24). Pooling took paper 2b's interval from [-0.040, 0.343] on test-A to [-0.067, 0.204], a width of 0.271 against 0.383, with the centre still near zero.

4.2 The forecast price

Table 6: The model coefficient when the market term is the pre-race racecard forecast price. The pooled row is reported because Section 5 uses it; the halves differ, so it is an average of two different figures rather than a single one.

Model	Sample	Races	β_{mod}	SE	95% CI	z
Paper 2b	Pooled	2,182	0.893	0.072	[0.751, 1.034]	12.35
Paper 2b	Test-A (earlier)	1,095	1.037	0.103	[0.835, 1.238]	10.09
Paper 2b	Test-B (later)	1,087	0.746	0.102	[0.547, 0.945]	7.33
Paper 3 (GBT)	Pooled	2,182	0.872	0.068	[0.738, 1.005]	12.81
Paper 3 (GBT)	Test-A (earlier)	1,095	1.026	0.098	[0.833, 1.219]	10.42
Paper 3 (GBT)	Test-B (later)	1,087	0.723	0.094	[0.539, 0.908]	7.68

Against the forecast price the smallest of these coefficients is more than ten times the pooled starting-price figure, and every z exceeds 7. Adding the model term to a market-only model gives likelihood-ratio statistics of 104.6 and 112.9 on one degree of freedom on test-A, 54.6 and 60.2 on test-B.

The halves differ ($p = 0.044$ and 0.026), so there is no single figure for the forecast price.

The archived forecast price gives 0.884 on test-A for paper 2b, between the two arms above and much nearer the pre-race forecast.¹

Both models return the same answer on both prices, despite disagreeing on the top horse in about a third of test races (paper 3, §4.1). Two different views of the same features are priced alike.

5 Discussion and limitations

Papers 1 to 3 could not tell a model the market dominates from one that is partly complementary. On the starting price it is the first: everything the model’s features carry is, so far as this construction detects, already in the price. That is bounded rather than absolute — a contribution up to 0.204 sits inside the interval. What the result rules out is a contribution of the size the same models produce against a forecast price.

5.1 Limitations

The forecast price is a published opinion, not a traded price. It is printed in a racecard, no money stands behind it, and nobody is exposed if it is wrong. Fitted alone its market coefficient is 0.885, below 1, meaning the likelihood wants it flattened — a price making sharper distinctions between runners than outcomes justify. That is measurement error, and it leaves signal the market should have claimed for the model term to pick up. So arm A measures the model against a published opinion, not against a market. The trading-relevant quantity is the model’s value against an early *exchange* price, traded and money-backed. It lies between the two figures this paper reports — 0.068 against the settled starting price and 1.037 against the forecast price — and Smartform holds no exchange prices at any timestamp, so this dataset cannot place it.

A large coefficient against a weak price is not the model approaching the market. The starting price alone reaches a McFadden pseudo- R^2 of 0.1378 on test-A; the best blend built on the forecast price reaches 0.0774, and the forecast price alone 0.0526. Adding the

¹Arm C’s coefficients are 0.884 and 0.896 on test-A, 0.620 and 0.617 on test-B, and 0.757 and 0.756 pooled. The gap between arm A and arm C bounds the errors-in-variables inflation in arm A, and is used for that purpose in Section 5.

model to the forecast price helps a great deal and still leaves it well short of the starting price with no model attached.

The size of the measurement error is not known. The archived forecast price is cleaner than the pre-race one, and possibly also later (Section 7). The gap between the two arms — pooled, 0.136 and 0.116, about 14% of arm A — therefore shows that arm A’s coefficient is too high, without fixing by how much.

One racing code, one surface, one decade. UK All-Weather Flat handicaps at four courses, 2006 to 2015. Nothing here establishes the result holds for turf, for jumps, for another jurisdiction, or for today’s market.

6 Conclusion

Blended with the starting price on 2,182 held-out races, the model contributes 0.068 and 0.039.

That raises the bar for paper 5. Paper 3 pointed at a learned representation of a horse’s run history in place of hand-built summaries of it. If the settled market already prices what these features carry, compressing the same histories more efficiently produces a model that is already priced, however good the compression. The representation has to extract something the summaries discard *and* that the market has not itself inferred.

Anyone else using Smartform should note Section 7: the two forecast-price columns disagree on a third of rows, and the documentation gives no reason why.

7 Appendix A: Forecast-price provenance in Smartform

This appendix is for anyone else reusing Smartform. It records a divergence between two columns the documentation describes identically.

Smartform holds a forecast price in two places. `daily_runners` carries the forthcoming day’s racecards, available from the night before racing; `historic_runners` is the results archive, written the following day with the previous day’s results. On this machine the daily tables update at 00:30 and the historic tables at 05:30. Arm A of this paper takes the daily value, arm B the starting price from the archive, and arm C the archive’s own forecast price.

The Smartform manual (*Smartform Data Feed* 2011) documents both forecast columns identically, as the forecast price from the racecard. It gives no reason for the two to differ, and it predates the end of this paper’s data window by four years.

They do differ. On the 32,898 training-split runner rows carrying both values and a starting price, the two disagree on 32.5% — typically by one or two steps of the bookmakers’ fractional

ladder, 10/1 against 12/1 rather than anything wholesale. Where they disagree the archived value sits closer to the starting price: mean absolute log distance 0.395 against 0.448, with 59.0% of the 10,691 disagreeing rows moving toward it.

Fitted alone on the test universe, the archived value is the better predictor of the two.

Table 7: Each forecast-price column fitted alone as the market term, same races.

Column	β_{mkt}	SE	z	Log-likelihood	McFadden R^2
Pre-race (daily)	0.962	0.043	22.53	-4283.1	0.0601
Archived (historic)	1.064	0.044	24.13	-4238.9	0.0698

Carrying both in one conditional logit, the archived value adds to the pre-race one: a likelihood-ratio statistic of 95.6 on 1 degree of freedom against the pre-race-only model. The two columns correlate within race at 0.959, so that fit is identified off little residual variation and its individual coefficients are unstable — the pre-race coefficient turns negative (-0.375) while the same column fitted alone is strongly positive (0.962). Nothing here rests on that sign.

Two explanations remain live, and they bias β_{mod} in opposite directions.

- **A later racecard edition.** Forecast prices are revised between editions, the daily feed holds an earlier one and the archive a later one. Arm C is then contaminated by information that accrued after arm A was available.
- **The same price with transcription noise removed.** The daily value carries error, the archive is the corrected version, and the two differ by noise rather than timing. Arm A then carries an errors-in-variables attenuation that inflates β_{mod} .

The design does not separate them. Both predict exactly what is observed: a later price knows more, and a de-noised price also predicts better and sits closer to the starting price once the noisy one is conditioned on. The manual, documenting both as the same racecard price, gives no support for the first over the second. We do not claim to have resolved it, and Section 5 carries the consequence — that the arm A to arm C gap is an upper bound on the attenuation rather than an estimate of it.

For reuse: take the forecast price from `daily_runners` wherever a verifiably pre-race value is needed, expecting it to be absent before March 2008 and patchy for a year after (Section 3). The `historic_runners` column is written after the race and does not reproduce it.

8 Appendix B: Software and reproducibility

The software stack is the same as papers 1, 2a, 2b and 3’s, described in full in paper 1’s Appendix B: R, the tidyverse, {DBI}/{RMariaDB} for the Smartform database, {renv} for

package pinning, `{targets}` for the pipeline, and Quarto for typesetting. `{mlogit}` fits the blend directly, as it fits the conditional logits of papers 1 to 2b — `{tidymodels}` has no conditional-logit engine, and the same documented exception applies.

Three things are specific to this paper.

A separate pipeline and a separate store. Paper 4 runs from its own script, `_targets_p4.R`, into its own store, `_targets_p4`, rather than sharing the pipeline papers 1 to 3 use. Papers 2b and 3 are inputs here, and reading them from a pipeline that could also rebuild them invites a class of accident this paper cannot afford: nothing paper 4 does is allowed to alter a published result. The upstream targets are read from the main store read-only, and their content hashes are recorded in a `p4_upstream_fingerprint` target, so that if a paper 2b or 3 result ever changes, every paper-4 target downstream of it is invalidated rather than silently going stale.

The two are run separately:

```
targets::tar_make()                                # papers 1-3
targets::tar_make(script = "_targets_p4.R",        # paper 4
                  store  = "_targets_p4")
```

Unlike papers 1 to 3, this document does not call `targets::tar_config_set()` in its setup chunk. That function writes the project-root `_targets.yaml`, which papers 1 to 3's setup chunks also write; pointing it at the paper-4 store would send their next render at the wrong store. Paper 4 passes `store =` to each `tar_load()` call instead and leaves `_targets.yaml` alone.

Verification gates. Two standing checks guard this paper, in addition to the series' existing ones. `scripts/verify_p4_market_probs.R` asserts that paper 4's market-probability helper is the same arithmetic paper 1 §5.1 already applies to starting prices, by rebuilding the stored `win_market` column of paper 3's test predictions from raw prices and requiring agreement to 10^{-12} ; observed agreement is exact. It also checks that probabilities sum to one within race, that the construction is invariant to a uniform rescaling of prices, that the freshly pulled starting prices match the stored ones, and that the price join cannot fan out. `scripts/verify_p4_data_targets.R` ties Section 3's live targets to the standalone P4-0 audit that gated the paper, and asserts that the distributional summaries touch no test race.

What was fitted, and what was not. Papers 2b and 3 are not refitted anywhere in this paper; their stored test-set predictions are read as given. The only models fitted here are the blends themselves and their market-only references, each a one- or two-parameter conditional logit. The complete pipeline, including the database pulls, runs in well under a minute — which is the point made in Section 6 about the cost of this paper relative to the work it informs.

One departure from the series template is worth recording so it reads as deliberate. Papers 1 to 3 answer three questions — ranking, win-picking and betting value. This paper answers the first kind of question only and has no betting-value section, for the reason given in Section 1: a

blend that tracks the starting price bets into the starting price's own margin, so its backtest has a foregone conclusion.

Acknowledgments

I have been interested in modelling horse racing since first acquiring the Smartform database more than a decade ago. The project has been picked up and put down many times, the obstacle almost always being the gap between an idea and a running implementation. What is different now is that working alongside Claude Code (Anthropic's coding agent), with web-based Claude as a research companion, has compressed that gap dramatically. I am grateful to my brother, Michael Gillen, for the suggestion to build the model on All-Weather races — the foundation of the scope choices set out in paper 1. The design choices — dataset, scope, model, comparison, editorial line — are mine. The implementation, the bulk of the draft prose, and a great deal of the verification were Claude Code's, under my direction; every output was reviewed and edited before it reached the paper. I do not yet know what general lessons this experiment points to about the value of human–AI collaboration in applied statistics. That uncertainty is part of why I am doing the work, and these papers are partly a record of that experiment.

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